SOFTWARE REQUIREMENTS SPECIFICATION

AI Powered Style Trained Social Media Post Generator

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Status	Draft: For Engineering Review
Audience	Software Engineering Team

1. Purpose

This document defines the functional and nonfunctional requirements for an AI assisted content generation system that enables the marketing team to produce social media posts in the authentic writing style of specific, pre trained human writers without hallucinating facts, inventing data, or producing generic AI sounding content.

2. Problem Statement

The current approach of using general purpose AI tools (e.g., ChatGPT, Claude) to write social media posts produces content with the following unacceptable characteristics:

- Generic, recognisably AI generated wording that reduces audience trust and engagement.
- Hallucinated facts, quotes, or statistics not provided by the user.
- Inconsistent voice posts do not sound like the specific human writers the team wishes to emulate.
- Platform mismatch posts written for X (Twitter) may be too long or formal; Facebook posts may lack appropriate depth or structure.
- Low reach on X and Facebook as a direct result of the above.

3. Scope

The system is scoped to:

- Support a minimum of 4 named writer personas (e.g., Zack, Luke, and 2 additional writers to be defined).
- Support 2 target platforms: X (Twitter) and Facebook.
- Accept user supplied context as the sole factual basis for generated posts.
- Produce output that mirrors the vocabulary, tone, rhythm, and structural patterns of the selected writer's persona.

4. Writer Personas & Training Data

4.1 Persona Definition

Each writer in persona represents a real person whose public social media post will form the training dataset for that style. The system must replicate the following stylistic attributes per persona:

- Vocabulary range and word choices
- Sentence length and rhythm
- Use of punctuation, line breaks, and whitespace
- Tone (e.g., conversational, authoritative, humorous, direct)
- Post structure and opening/closing patterns
- Use of hashtags, mentions, or calls to action

4.2 Training Dataset Requirements

Attribute	Requirement
Minimum posts per persona	50 posts
Platform split	Posts sourced from the same platform the persona is known for
Data format	To be determined by engineering (raw text, JSON, CSV, etc.)
Data freshness	Posts must be recent and representative of the writer's current style

Data collection	Collected and verified by the marketing team before training

The marketing team is responsible for supplying, curating, and approving all training data for each persona.

5. Functional Requirements

5.1 Style Selection

- The user must be able to select a specific writer of persona (e.g., "Write in Zack's style").
- The user must be able to request all available styles simultaneously for comparison.
- The system must not mix styles across personas in a single output.

5.2 Platform Awareness

- The user must specify the target platform: X (Twitter) or Facebook.
- The system must apply platform appropriate constraints automatically (character limits, format, expected tone).
- The same brief must produce structurally different outputs depending on the selected platform.

5.3 Input Handling

- The user provides a brief: topic, key points, and any specific facts or data to include.
- The system must use only the information provided as a factual basis for the post.
- The system must not add external facts, statistics, quotes, or claims not present in the user's input.

5.4 Output

- The system must return a complete, ready to publish (or near ready) post.
- When multiple styles are requested, all outputs must be clearly labelled by persona.
- Outputs must not include AI disclaimers, preambles, or meta commentary.

6. Non-Functional Requirements

Attribute	Requirement
Hallucination Control	The system must not fabricate any fact does not present in the user's input. Zero tolerance.
Style Fidelity	Generated posts must be indistinguishable by a human reviewer familiar with the writer from authentic posts by that writer.
Response Time	Single style output delivered within a reasonable time (to be benchmarked during development).
Usability	The interface must require no technical knowledge. A non-technical marketing team member must be able to operate it without assistance.
Maintainability	New writer personas must be addable without system rebuilding. Only training data ingestion is required.
Accuracy	Content must faithfully reflect the user supplied brief without omitting or distorting key information.

7. Constraints

- Training data must be legally obtained, and the team must have the right to use it.
- The system must not store or expose training data in raw form to end users.
- Platform character limits (e.g., X post length) must be respected automatically.
- The solution must handle 4 personas and 2 platforms in its first release.

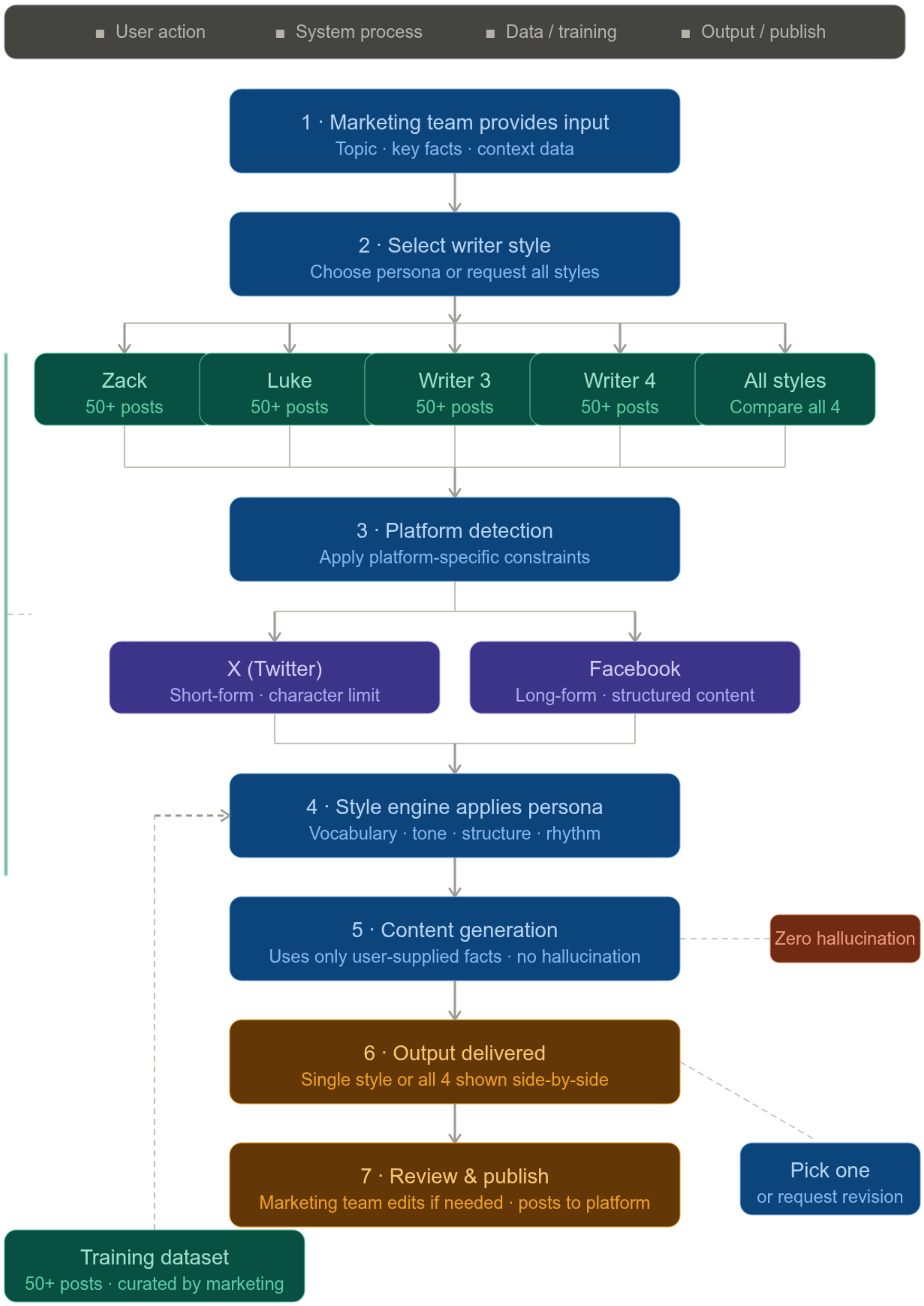
8. Assumptions

- The marketing team will supply and curate a minimum of 50 posts per writer persona prior to development.
- Each writer in person has a consistent, identifiable writing style across their posts.
- Users will provide sufficient context in their briefs for the system to generate accurate content.
- The 4 writers' personas and their respective platforms will be confirmed before development begins.

9. System Flow Diagram

The following table describes the end to end request flow, from user input through to published post.

Step	Stage	Description
1	**User Input**	Marketing team member provides topic/brief + context data + desired writing style (Zack / Luke / etc.) + target platform (X / Facebook)
2	**Style Selection**	User selects one of the trained writer personas OR requests all styles to compare
3	**Platform Detection**	System identifies target platform (X or Facebook) and applies platform specific constraints: character limits, tone, format
4	**Style Engine**	System retrieves trained style profile for selected writer. Applies vocabulary, sentence structure, tone, and posting patterns from training dataset
5	**Content Generation**	Post is generated using ONLY for the user provided context data. No fabricated facts, statistics, or claims beyond the supplied input
6	**Output**	System returns the generated post(s). If multi style was requested, all variations are shown side by side for user to pick
7	**User Review & Publish**	Marketing team member reviews, optionally edits, then publishes to target platform



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10. Open Questions for Engineering

1. What AI/ML architecture best satisfies the style of fidelity and zero hallucination requirements simultaneously?
2. How will style drift over time (if a writer's posting style changes) be managed?
3. Should the system support iterative refinement (user requests adjustments to a generated post)?
4. What is the minimum viable dataset size that produces acceptable style fidelity, validated by the marketing team?
5. What interface will the marketing team use: a web app, chat interface, or integrated tool?

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